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Palaeomagnetism of Sukinda chromites and their geological implications

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Summary. Thirty-eight oriented samples were collected from seven sites over the Precambrian Sukinda chromite ore (between 85°43′E; 21°01′N and 85°50′E; 21°03′N), which is in the form of a plunging syncline. On isolating the primary component of magnetization and affecting the necessary tectonic correction, four sites (A, B, C, F) were found to possess a reverse magnetic polarity ($D = 210°$, $I = +22°$, with $K = 122$ and $\alpha_{95} = 6°$) and one site (E) was found to bear a normal polarity ($D = 49°$, $I = -16°$ with $K = 474$ and $\alpha_{95} = 2°$). The remaining two sites D and G were rejected, the former because of scatter in directions and the latter due to overprinting by a dominant secondary magnetization, induced by lateritization. On comparison of VGPs evaluated for these chromite deposits, with those of other Precambrian rocks almost of a similar age, it is seen that these chromites are probably older than the Gwalior Traps whose age is 1830 ± 200 Ma. From the rock magnetic and microscopic examination it is deduced that the massive chromites of the Sukinda valley have not been affected magnetically to any great extent by serpentinization, though their host rocks are serpentinized ultramafics. Consistent directions of remanent magnetism point to the fact that these deposits crystallized in fairly stable conditions, characteristic of the stratiform type of chromite deposits, therefore, leading us to infer the Sukinda chromites to be of that type. These palaeomagnetic data, when subjected to the fold test, show the primary remanence to pre-date the tectonic disturbance and hence prove these chromites to have consolidated prior to the regional tectonic activity of the area.

Introduction

Sukinda valley contains the largest deposit of chromite in India. Geological investigation of these deposits have been carried out from time to time by various workers, pioneered by Deekshitulu & Perraju (1955), followed by Das Gupta (1959), Sahu & Bagchi (1959), Mitra

(1962, 1974), Acharya (1964), Prasad Rao, Rao & Deekshitulu (1964), Banerjee (1972), Chakraborty (1972, 1973), Chakraborty & Majumder (1976), Sahoo & Van Der Kadden (1976) and Chakraborty, Chakraborty & Majumder (1980). Thus, substantial geological knowledge of the valley has been accumulated from the investigations carried out over the last couple of decades. In spite of this, controversies still exist among various workers over the origin, type and age of these chromites.

Keeping these controversies in mind, detailed geological, geochemical, rock magnetic and palaeomagnetic investigations have been taken up. Of these the palaeomagnetic results are presented here.

Geology

The chromites of Sukinda valley occur in a zone of ultramafics which is intrusive into quartzites of the iron-ore series. The chromite occurs in thick layers that might have been thickened by the folding that affected them. These ores appear to be stratiform in an ultramafic rock. Here, the chromite layers have a strike length of about 12 km in a NE–SW direction. The deposit extends from Katpal (85° 43′: 21° 1′) in the west to Saruabil (85° 50′: 21° 3′) in the east. The area has been affected by major diastrophic movements which produced a SW-plunging syncline. This involves both the chromitiferous ultramafics and the enclosing rocks (Precambrian quartzites and banded iron formations).

The stratigraphic succession of rocks in the Sukinda valley as proposed by Chakraborty *et al.* (1980) is given in Table 1. The dips of the chromite layers lie between 35° and near vertical, with the differences in dip being due to post-depositional deformation.

From the various structural findings, Chakraborty (1972) emphasizes that the chromites of the Sukinda are true stratiform deposits, even though the continuity of the individual beds has been destroyed or confused by later earth movements. Contrary to this suggestion, Chatterjee & Banerjee (1964) conclude that the Sukinda chromite deposits do not conform to either the stratiform or alpine types of Thayer (1960). They postulate the chromite-bearing ultramafics as small, disconnected podiform bodies arranged in an accurate pattern. However, it is generally agreed that both stratiform and podiform chromites occur in India.

The time of emplacement of the Sukinda chromite deposits as assigned by Chakraborty *et al.* (1980) is earlier than that of the Kolhan deposit which belongs to the early Proterozoic

Table 1. Stratigraphic succession of the Sukinda valley (Chakraborty *et al.* 1980).

Alluvium
Laterite (nickeliferous in places)
Dolerite
Granite and granite gneisses
Orthopyroxenite (enstatolite)
Ultramafic intrusion with chromite
Sandstone (quartz-arenite): Kolhan Group (?)
Conglomerate
---------- Unconformity ----------
Iron ore group
  Banded iron formation
    with interbedded volcanic tuff
    and alteration products
  Quartzite with intraformational conglomerate
--------Basement not exposed---------

period. Apparently all chromite deposits in India are either early or Middle Precambrian in age, more probably the former, since they have been affected by pre-Algonkian orogenies (Chakraborty *et al.* 1980).

Sampling

Samples were collected from fresh exposures of the chromites in mining quarries all along their total strike length of 12 km. Sampling was carried out on both the southern and northern limbs of the plunging synclinal chromite bands. Three to 10 oriented block samples were collected at each of seven sites (Fig. 1), one site at each of the Sukerangi (A), TISCO grade I (B), TISCO grade III (C), Kaliapani (D), Kalarangi (E) and Katpal (F) quarries on the southern limb of the syncline and one site at Pentasai (lateritized chromite, G) on the northern limb. On the southern limb though active mining is in progress, in many more quarries than the ones mentioned above, sampling for palaeomagnetic purposes in these other quarries was not possible as the ore found there is of a friable nature. On the northern limb, sampling from more than one site was not possible due to lack of suitable exposures.

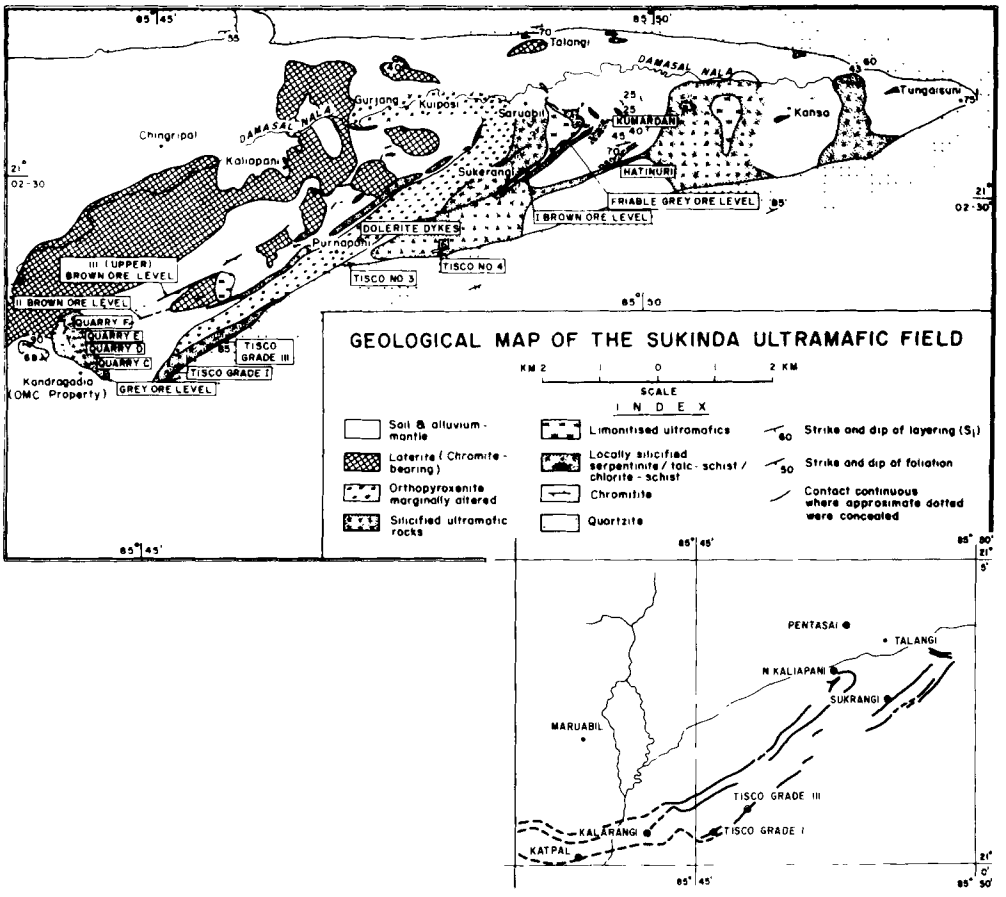


Figure 1. At the top of the illustration is a geological map of the Sukinda valley, Orissa. To the bottom right is a map showing the locations of the chromite bands (identified with the aid of bore hole information). Sampled sites are marked.

Laboratory methods and data analysis

Block samples were cored and cut in the laboratory. A minimum of three cylindrical specimens of 2.5 cm diameter and 2.5 cm length were obtained from each block sample resulting in a total of 105 specimens. Remanent magnetism of these specimens was measured on an astatic magnetometer. The NRM intensity (J_n) varies between 9.5 and 109×10^{-3} A m^{-1}. The mean NRM direction for each site and the associated statistical parameters are presented in Table 2.

Table 2. The mean NRM directions for the sites studied and their statistical parameters.

Site	N	n	NRM D_m°	NRM I_m°	K	α_{95}°
A	3	7	132	−13	26	11
B	5	13	173	−42	25	8
C	5	15	172	−36	36	6
D	10	22	Scatter	–	–	–
E	5	13	4	−9	156	3
F	4	12	172	−29	290	2
G	6	23	356	+34	42	5

N = number of samples.
n = number of specimens.
D_m = mean declination.
I_m = mean inclination.
K = Fisher's precision parameter.
α_{95} = radius of circle of confidence at 95 per cent probability level.

Alternating field (AF) demagnetization

To test the stability of the magnetic carriers and to isolate the primary component of magnetization, AF demagnetization was performed using an apparatus similar to that described by Creer (1959). Thirteen specimens were subjected in a successive step-wise fashion up to the maximum peak field of 120 mT. Curves of normalized intensity against demagnetizing field exhibit a sharp drop at fields less than 20 mT (Fig. 2) revealing the existence of a high percentage of soft magnetization in specimens from all sites except D. In spite of the large loss in intensity (85–95 per cent) by 120 mT the direction of magnetization is maintained right from the NRM stage (Fig. 3) except for the case of site A in which the vector was slightly astray initially, but migrated to settle with the rest during incremental demagnetization. In addition to these pilot specimens, demagnetization of two specimens per sample at peak fields of 10, 20 and 30 mT was undertaken to provide evidence of consistency.

Thermal demagnetization

Since the magnetic mineral exhibits an exceptional directional stability upon AF demagnetization and, as stated earlier, some remanence still remains after AF treatment at 120 mT especially in specimens from site D, the presence of haematite, one of the major magnetic minerals in rocks, which can have very high coercivities (Roguet 1954) cannot be ruled out. Hence thermal demagnetization was applied to one pilot specimen from each site by heating

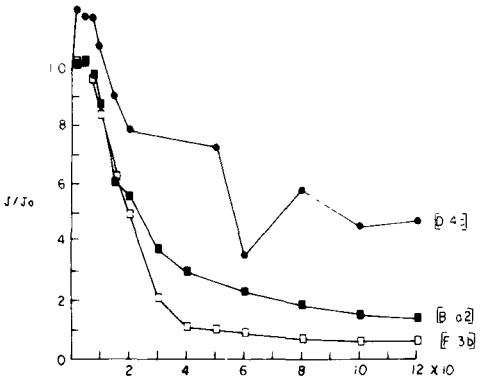


Figure 2. Decay of normalized intensity of AF demagnetization for typical samples; peak fields in mT. Site and sample numbers are given in brackets.

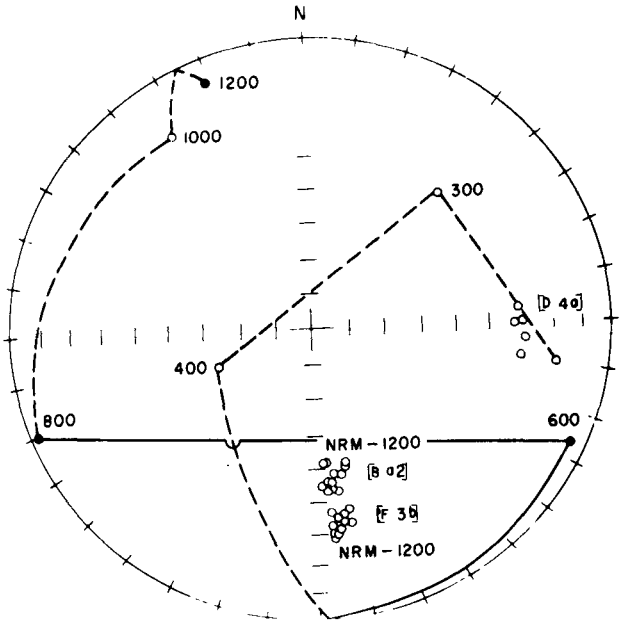


Figure 3. Equal area projection showing directional changes during AF demagnetization for typical samples; peak fields in mT. Solid circles indicate positive dips, open circles negative dips. Site and sample numbers are given in brackets.

first to 100°C, then in steps of 100°C up to 400°C and subsequently in steps of 50°C up to a maximum temperature above which either the intensity became too weak for adequate measurement or major chemical changes appeared to have taken place. When heated to a temperature of 300°C and beyond, the directional stability of the magnetic vector was lost. As this behaviour could possibly be attributed to experimental errors, a basalt test specimen with induced TRM (cooled from 600°C in the Earth's field) was also treated along with the batch. Contrastingly the magnetic direction of the test specimen was faithfully preserved and carried along up to the Curie temperature (T_c) (Fig. 4). Intensity decay during thermal demagnetization is illustrated in Fig. 5.

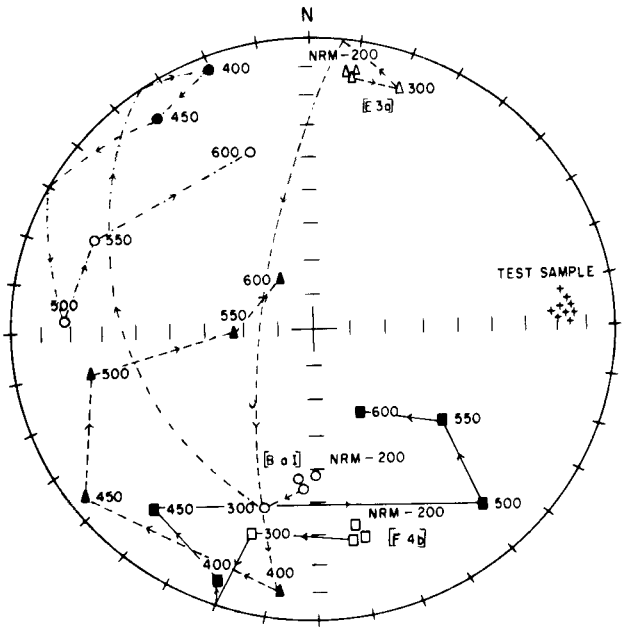


Figure 4. Equal area projection showing directional changes for typical samples as a result of thermal demagnetization at various temperatures (°C) in zero field. Open symbols indicate negative dips, solid symbols positive dips.

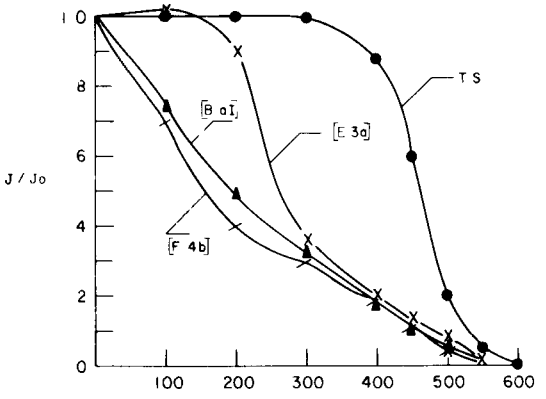


Figure 5. Decay of remanent intensity upon thermal demagnetization.

IRM tests

These studies were initiated with a view of ascertaining the iron oxide responsible for the remanence and also to understand its chemical behaviour upon heating. IRM acquisition experiments show the specimens to saturate between 60 and 80 mT and to exhibit a coercive force (H_c) between 10 and 20 mT (Fig. 6). In none of these samples was there any further increase in intensity up to a maximum field of 700 mT (Fig. 6). These IRM experiments suggest that the dominant magnetic carrier present is magnetite or a mineral with similar properties and that haematite or goethite are absent.

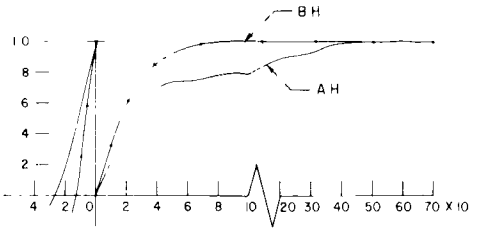


Figure 6. Normalized IRM acquisition and demagnetization curves, fields in mT. BH indicates before heating and AH after heating.

The rock magnetic evidence presented here, though not conclusive, is consistent with the dominance of magnetite or magnetite-like material. Secondly the magnetization is of a thermal nature. Detailed rock magnetic experiments are being conducted separately to identify the magnetic carrier in these chromites.

Results (palaeomagnetic)

From the comparison of vector migration and intensity decay curves (Figs 2, 3, 4 and 5) both upon af and thermal demagnetization, it is seen that the vector obtained by the former is more reproducible and stable. On the contrary, the magnetic vector during thermal demagnetization beyond 300°C is erratic mainly due to chemical changes. This was tested by a comparison of IRM curves, constructed before and after heating at 400°C (Fig. 6). There is an increase both in H_c and J_s for the heated samples. Such a behaviour is mainly attributed to change in the chemistry of the magnetic mineral. Therefore, for bulk cleaning of these specimens AF demagnetization was preferred. Optimum fields of 10, 20 and 30 mT were selected, based on the information furnished by the pilot intensity decay and vector migration data. Statistical parameters were computed giving unit weight to each sample as suggested by Fisher (1953). Results showing minimum dispersion are reported in Table 3. As given in this table, remanent magnetic vectors from sites A, B, C and F show reverse polarity while that from site E shows normal polarity. The mean direction of remanent magnetism for the five sites put together (after tectonic correction) is $D_m = 38°$, $I_m = -21°$ ($K = 58$, $\alpha_{95} = 8°$). Specimens from site D showed scattered NRM directions which could not be improved either by AF (Fig. 3) or thermal cleaning (Fig. 4).

Samples collected at site G are the lateritized chromites. These, when subjected to palaeomagnetic and rock magnetic tests (not detailed here) appear to be dominated by a magnetic component formed during lateritization. Therefore, this site is not taken up here for further description.

Fold test

On effecting the necessary tectonic corrections the site-mean magnetic directions show an appreciable change resulting in an approximately antiparallel direction. This correction results in a marked improvement in the Fisherian (1953) statistical parameters either on taking all the five sites (before correction $K = 10$ and after correction $K = 58$) into consideration or just the four reversed ones (before correction $K = 72$ and after correction $K = 122$). An improvement in the precision parameter (K) indicates a positive fold test. The number of sites in this study is too small to allow a rigorous examination of the statistical significance of the fold test result, but it seems probable that a substantial portion of magnetization pre-

Table 3. Mean site directions before and after application of tectonic correction, and palaeomagnetic pole positions for data after tectonic correction.

Site	Treatment (mT)	Before tectonic correction				After tectonic correction		Pole position	
		D°_m	I°_m	K	α_{95}	D°_m	I°_m	Lat	Long
A	20	161	−28	79	7	218	+23	40 N	214 E
B	30	166	−40	41	7	205	+28	48 N	228 E
C	30	175	−45	43	6	206	+16	51 N	222 E
E	30	4	−10	474	2	49	−16	34 N	202 E
F	10	172	−30	643	2	210	+22	46 N	221 E

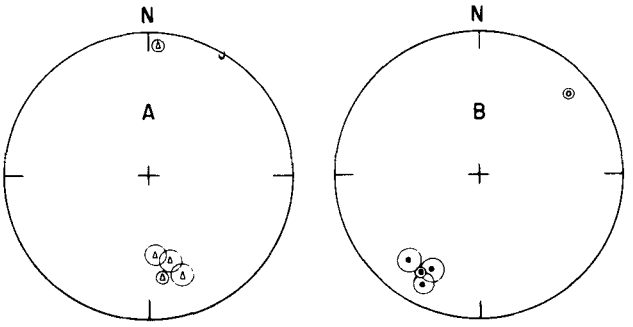


Figure 7. Stereographic plot of the stable magnetic directions (site means) before (triangles) and after (circles) tectonic correction. Circles represent 95 per cent confidence limits.

Table 4. Statistical parameters for Sukinda chromites before and after application of tectonic corrections.

No. of sites	Before K	Before α°_{95}	After K	After α°_{95}
5	10	20	58	8
4	72	8	122	6

Notations used as in Table 2.

dates the tectonic activity of the Sukinda valley. (Correction for individual samples was not attempted as dips within sites do not vary.) Directions of remanent magnetism before and after correction are given in Fig. 7 and the associated statistics in Table 4.

Discussion

Experiments detailed above show the characteristic component of magnetization to possess directional stability under AF demagnetization. The mineral responsible could be magnetite. Absence of free haematite in magmatic chromites (similar to the ones studied in this investigation) has been verified in the USGS laboratories (Banerjee 1972). However, AF demagnetization of NRM shows that most of the remanent magnetization (85–95 per cent) is removed in fields below 120 mT, for samples from all sites except the anomalous site D. Thus, it seems that haematite or goethite which are usually characterized by very high coercivities, even if present, do not play an important role in the NRM of the bulk of the samples. Hence, palaeomagnetic directions represented by these chromites are likely to be primary and pre-tectonic in age.

Effects of serpentinization

Since the chromite deposits of the Sukinda valley lie in serpentinized ultramafics, it could be suspected that the magnetic minerals present are a product of serpentinization. During serpentinization the iron which is originally present in olivine or pyroxene redistributes and enters into solid solution in serpentine minerals or brucite or forms an opaque phase which is mostly magnetite. The common alteration assemblage produced by serpentinization of an ultramafic rock is lizardite, crysotile, magnetite, brucite and antigorite (Moody 1976). If alteration during serpentinization was sufficient to produce enough magnetite, the primary direction of magnetization should be overprinted by a secondary one representing the time of serpentinization. However, during both AF and thermal treatment no such magnetization was observed, the simple reason being that within the massive type of chromite ore (all the samples for palaeomagnetic purpose were collected from massive ores) the degree of alteration was not enough to precipitate appreciable amounts of magnetite (microscope studies on thin and polished sections confirm this). This strongly suggests that the stable component of magnetization was acquired during the primary consolidation of the chromites and that serpentinization has not had any significance effect on the magnetic mineralogy of these deposits.

Prototectonics and RM directions

During the consolidation of chromite ore deposits, intensive tectonic movements may take place, especially during the later stages of the early crystallization processes. These tectonic movements or prototectonics are magmatic movements of chromite crystal masses in a still-liquid silicate matrix. These are most effective in the development of the so-called 'alpine type' chromite deposits (Thayer 1960, 1962). On the other hand, a world-wide review of chromite deposits shows that the stratiform and layered types are associated with only very weak prototectonics (Bushveld, Great Dyke, Stillwater etc.) and hence are not tectonically disturbed during crystallization. It is generally accepted that TRM is acquired when the magma cools through the Curie temperatures of the various magnetic minerals contained in it, in the presence of the geomagnetic field. The alignment of the magnetism of these minerals in a uniform direction is achieved to the utmost satisfaction when the cooling/consolidation takes place in stable tectonic conditions. Hence, persistent magnetic directions among samples and sites for these chromite deposits could be attributed to the almost negligible prototectonics involved during crystal settling of the Sukinda chromites.

The positive fold test also suggests that these chromite exposures, which appear as isolated pockets of ore, were originally more continuous, but the alignment has been destroyed by later earth movements. Therefore, from these two pieces of palaeomagnetic evidence, it could be inferred that the chromite deposits of the Sukinda valley are of the true stratiform type of Thayer (1960).

Age of Sukinda chromites

The Precambrian polar wandering curve for India is not very reliable due to the lack of critical age (radiometric) data, hence, comparing the virtual geomagnetic pole (VGP) for the Sukinda chromites with such a path to infer their age may result in uncertainty. The time of emplacement of these chromites can probably be estimated by a comparison (Fig. 8) of the VGP for these rocks with others which fall in close proximity. Five poles (Lower Gwalior Traps, GT, Klootwijk 1974; Kolar dykes, KD, Reddy & Prasad 1979 and Hyderabad dykes, HD, Verma, Pullaiah & Hasnian 1968 and Kumar & Rao 1979) which are nearest to the VGP

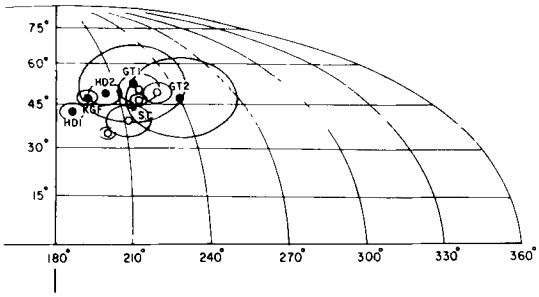


Figure 8. Molleweide's projection showing the virtual geomagnetic pole positions for the Sukinda chromites (five sites), indicated by open circles, star represents mean for all the five sites, solid circles indicate Gwalior Traps (GT), Hyderabad dykes (HD), and Kolar dyke (KD).

of the chromites are taken. All the above-mentioned igneous rock suites are considered to be Precambrian with one age (Rb/Sr whole rock) dating by Crawford & Compston (1970) at 1830 ± 200 Ma for the Gwalior Traps. This study, therefore, suggests that the Sukinda chromites may be more or less equivalent in age to the Gwalior Traps.

A comparison of the regional structural patterns of the Eastern Ghats Orogenic Belt, with the structure of the Sukinda valley shows these two units to have their respective main fold axes parallel to one another (Narayanaswamy 1966). Therefore, this orogeny which took place around 1570–1625 Ma (Sarkar *et al.* 1964) has left its impression on the pre-existing chromite-bearing ultramafics. The positive palaeomagnetic fold test is in agreement with this structural implication and indicates that the chromites are probably older than the orogeny (1570–1625 Ma).

Conclusions

(1) The magnetic mineral responsible for carrying the primary remanence in the Sukinda chromite ore is highly stable and is most probably magnetite.

(2) The effect of serpentinization (as seen from both microscopic and magnetic examination) is not sufficient in the massive type of chromite deposits to leave an impression on the pre-serpentinization remanence.

(3) Estimation of the vigour of prototectonics involved with the aid of palaeomagnetic data (consistent palaeomagnetic directions) suggests that the chromite deposits of the Sukinda valley were associated with only weak prototectonics (which is characteristic of the stratiform type of deposits). This supports the contention that these deposits are of a stratiform nature.

(4) Palaeomagnetism, by way of a fold test, indicates that the chromite probably consolidated prior to the regional folding of the area, associated with the Eastern Ghats Orogeny about 1570–1625 Ma, which is therefore the minimum age of the chromites. Comparison of the pole position for the Sukinda chromites with five other Precambrian poles suggest these ores may be roughly contemporaneous with the Gwalior Traps.

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